APEX CAPITAL MANAGEMENT & QUANT RESEARCH

AIMA Institutional Due Diligence Questionnaire (DDQ)

CME Globex Multi-Commodity Microstructure & Systematic Alpha Mandate

Mandate Name:	Apex CME Globex Quantitative Futures Alpha	**Benchmark:**	COMEX Gold Active Futures (GC / MGC)
Target Asset Classes:	CME Metals (GC, SI, HG), Energy (CL, NG), Rates (ZN)	**Target Prime Capacity:**	**$50,000,000 USD** (Prime Broker SMA / Fund)
Execution Gateway:	CME iLink3 / FIX 4.4 Co-Located Port :9800	**Risk Standard:**	**SEC 15c3-5 Pre-Trade Risk (< 35µs)**
Model Architecture:	Sparse MoE + DeepLOB (89.65% OOS) + Macro TFT	**Audit & Clock Sync:**	CFTC Rule 1.31 / MiFID II RTS 25 Microsecond

Pillar 1.0: FIRM OVERVIEW, GOVERNANCE & KEY PERSON PROFILE

1.1 Fund Structure & Governance

Apex Capital Management operates as a specialized quantitative research studio focused on high-frequency microstructure modeling and systematic commodities trading. Capital allocations are structured as Separately Managed Accounts (SMAs) via Tier-1 Prime Brokers (Interactive Brokers, Marex, StoneX) with independent daily clearing and custody.

1.2 Key Personnel & Experience

The research and execution desk is led by Principal Quant Architect Bhaskar Pandey, combining 10+ years of institutional machine learning, distributed systems architecture, and market microstructure modeling. Direct WhatsApp Institutional Allocator Desk: +91 9470778244 | Direct Contact: bhaskarauthor@gmail.com.

1.3 Capacity Constraints & AUM Caps

To preserve high Sharpe ratios (>2.40) and prevent market impact on COMEX order books, total strategy capacity is strictly capped at $50,000,000 USD across all SMA prime mandates.

Pillar 2.0: INVESTMENT STRATEGY & EMPIRICAL ML ARCHITECTURE

2.1 Model Architecture (Sparse Mixture-of-Experts)

The platform routes signals through an 8-expert Sparse Mixture-of-Experts (MoE) engine. An autonomous Softmax gating router assigns top-2 expert weights per microsecond tick, maintaining gating entropy of 1.327 nats (well above the 1.20 floor) and an SVD condition number $\kappa(W) = 4.11$, preventing expert collapse.

2.2 DeepLOB Empirical Benchmark Performance

The intraday order book model employs a spatial Inception-CNN + BiLSTM + Attention architecture trained on over 130,000 nanosecond MBP-1 order book snapshots. Empirical out-of-sample (OOS) directional accuracy achieves 89.65% across 229,104 trainable parameters, utilizing Marcos López de Prado Triple Barrier labeling (+1 profit target: 88.2% prec, -1 stop loss: 89.1% prec).

2.3 Macro Temporal Fusion Transformer (TFT)

A multi-horizon TFT model operates across 10 years of continuous 1-hour commodity bars (57,258 bars). Out-of-sample directional accuracies are audited at 50.92% (1h), 51.06% (4h), and 52.67% (24h) with Quantile Pinball losses (q10: 0.0418, q50: 0.0612, q90: 0.0425), capturing macro yield-curve and DXY cointegration shifts.

Pillar 3.0: EXECUTION INFRASTRUCTURE & CO-LOCATION

3.1 Order Routing & Exchange Connectivity

Direct CME Globex connectivity is maintained via iLink3 and FIX 4.4 TCP gateways operating on dedicated high-speed interfaces. Sub-millisecond order serialization translates model alphas into actionable limit and pegged orders with zero intermediate broker hops.

3.2 Pre-Trade Risk Firewall (< 35µs Latency)

All outbound orders are intercepted by an in-line SEC 15c3-5 compliant pre-trade risk firewall running in C++ / Cython with sub-35µs latency. Checks include: (1) Fat-finger size limits (max 25 lots), (2) Notional exposure caps ($1.0M), (3) 100 ms message throttling, (4) ±1.5% dynamic price collars, and (5) Limit-Up/Limit-Down (LULD) circuit breaker detection.

3.3 Zero Cloud IP Leakage Policy

100% of trading signals, weights, proprietary features, and microstructure order book tensors reside in secure, isolated bare-metal environments. No proprietary models or trading logic are ever exposed to public commercial LLM APIs.

Pillar 4.0: REGULATORY COMPLIANCE, CFTC & MIFID II AUDIT TRAILS

4.1 CFTC Rule 1.31 & Rule 575 Pre-Trade Compliance

The platform implements an immutable dual-write audit blotter. Every order request, pre-trade risk evaluation, and execution report is dual-written to local SQLite stores and an enterprise cloud Neon PostgreSQL cluster with nanosecond-resolution UTC timestamps.

4.2 MiFID II RTS 25 Microsecond Clock Synchronization

All trading events and FIX protocol drop copies feature microsecond-synchronized UTC timestamps (Tag 52=SendingTime, Tag 60=TransactTime), satisfying MiFID II RTS 25 regulatory requirements for algorithmic trading records.

4.3 SOH-Delimited FIX 4.4 Session Export

Compliance officers can download complete, authentic SOH-delimited (■) FIX ASCII audit logs directly from the platform or via authenticated REST endpoints (/api/v1/fix/audit-export) for automated reconciliation.

Pillar 5.0: LAKEHOUSE, DATA GOVERNANCE & CROSS-ASSET GAT

5.1 DuckDB 2.0 OLAP Microstructure Lakehouse

The platform maintains 10 years of continuous multi-scale market data (ticks, 1s, 1m, 1h, 1d) stored in date-partitioned columnar Apache Parquet. A zero-copy DuckDB OLAP engine enables sub-millisecond analytical aggregations across hundreds of millions of rows.

5.2 Cross-Asset Graph Attention (GAT) & Granger Latencies

A Spatial-Temporal Graph Attention Network continuously monitors 6 core macro assets: Gold (GC), Silver (SI), Crude (CL), Copper (HG), 10Y Yields (ZN), and US Dollar Index (DXY). Sub-second Granger causality establishes that DXY leads Gold by 320ms and 10Y Yield leads by 450ms, while Gold leads Silver by 180ms.

Pillar 6.0: OPERATIONAL DUE DILIGENCE (ODD) & DISASTER RECOVERY

6.1 Business Continuity & Failover (BCP/DR)

System architecture features zero-single-point-of-failure redundancy. The API and decision daemons are monitored by an autonomous 600-minute watchdog supervisor. Failover recovery time objective (RTO) is < 30 seconds, and recovery point objective (RPO) is 0 seconds via transactional WAL logging.

6.2 Real-Time Webhook Lead & Risk Alerts

In the event of an SEC 15c3-5 risk collar breach, Hawkes VPIN toxicity spike (>0.50), or allocator pilot registration, automated webhook dispatchers instantly broadcast push alerts to desk personnel via Telegram/Discord/REST endpoints and SMS/WhatsApp.

Pillar 7.0: PRIME BROKERAGE, CUSTODY & MARGIN STRESS TESTING

7.1 Custody & Prime Broker Relationships

Institutional accounts maintain direct custody of all assets through their clearing prime broker (e.g., Interactive Brokers LLC, member SIPC/FINRA). Apex operates strictly as a sub-advisor or trading signal provider via FIX drop-copy / CTCI feeds.

7.2 SPAN Margin Efficiency & Hedging Matrix

Utilizing cross-commodity statistical arbitrage and zero-cost options collar overlays, SPAN initial margin requirements are reduced by 72.9% relative to naked directional futures positions, maximizing capital efficiency for institutional SMAs.

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